SKITII CDSCO / IEC 62304 PACK

V&V As-Built vs Blocked

What the Yenepoya Trial app actually implements versus what SKT-TC-HR, SKT-TC-HRV and related execution sheets must not be pre-filled until PI / Quality freeze the operators.

VERSION
1.0

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CUSTOMER
Internal (Mindful Gurukul / CDSCO pack)

SCOPE
Map the 03-Sep-2026 V&V Word pack against the current Yenepoya Trial app

Control note

This document is **as-built engineering fact** from the current Trial stack. It is **not**:

- a completed Software Verification & Validation Report
- permission to pre-fill PASS / FAIL / Evidence IDs in `SKT-TC-*` sheets
- an approved SRS/SDS freeze of algorithm operators

The Word pack itself says: do not pre-fill PASS; expected results must be approved **before** observing Skitii output; unresolved numeric operators remain OPEN.

Use this file to decide **what testers can run later**, and **what PI / Quality must freeze first**. Do not copy these tables into execution Actual-result cells.

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1. Labelled configuration this map uses

The CDSCO pack describes Trial 1: dedicated Android tablet or dedicated device, Polar H10 over BLE, 20-minute two-phase session, four music categories.

Code used for this map

Layer	Repo	What exists
Session / BLE / HR / HRV / music	`Skitii-Trials-Yenepoya-HNF-APP`	Polar H10 BLE, `hrv.ts`, `discreteBands.ts`, 20-min session, OTP + PIN
Trial APIs	`Skitii-Trial-Backend`	Patients, sessions, readings, week-0
Trial Admin	`Skitii-Trial-Admin-Frontend`	Patient list, weekly forms, music, dashboard

Out of scope for this pack (same as SKT-SVVR-001)

- `Skitii-Desktop-Ble` (Pi / station kiosk engineering track)
- `Skitii-Product-Frontend` (commercial product, not Trial 1 labelled config)
- Global (cross-patient) ranking — Trial 2
- Computer-vision / facial-emotion modules — not in the product

The lifecycle draft still mentions React + Python bleak + Capacitor BLE. The Trial app is **React Native CLI + react-native-ble-plx**. SDS / ARCH should be corrected to that before freeze.

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2. Pack status (all 19 files)

Document	ID	Fill as executed V&V?	What this note records
Architecture diagrams	SKT-ARCH-001 v0.2	No (draft)	Stack mismatch vs Trial app
Cybersecurity protocol	SKT-CYB-001 v0.1	No	Auth/TLS exist; kiosk / AES-256 session store / SBOM do not
Harmonization register	SKT-HARM-001 v0.1	Process only	OPEN-01 … OPEN-11 still block release
Lifecycle plan	SKT-SDLC-001 v0.2	Process only	Draft; Class B still unapproved
Log templates	SKT-LOGS-001 v0.2	Empty by design	Add rows only for real DEF- / EV- / tagged builds
Risk / FMEA	SKT-RMF-001 v0.2	Process only	Residual risk after V&V, not now
RTM	SKT-RTM-001 v0.2	Mapping only	Evidence / Status columns stay empty
SDS	SKT-SDS-001 v0.1	Draft design	Operators still OPEN in the SDS itself
Software safety class	SKT-SSC-001 v0.1	Process only	Formal IEC 62304 class not approved
SRS	SKT-SRS-001 v0.2	Draft requirements	Many TBC (timeout, latency, AES-256, kiosk)
SVVR	SKT-SVVR-001 v0.2	**Do not submit as final**	Unexecuted report template
V&V plan	SKT-VVP-001 v0.2	Approve before any formal run	Not an evidence file
V&V execution master	SKT-EXEC-MASTER-001	After family sheets	Cover sheet only
BLE execution sheets	SKT-TC-BLE-001	After Polar run on frozen build	8 of 10 runnable in principle; 2 blocked
E2E execution sheets	SKT-TC-E2E-001	After frozen labelled config	~9 of 12 have product behaviour; 3 blocked
HR execution sheets	SKT-TC-HR-001	As-built PASS	Contiguous HR bands; 84.5 Restorative
HRV execution sheets	SKT-TC-HRV-001	Mixed	007–009 PASS after RMSSD alignment; 001–003/011 still blocked

Document	ID	Fill as executed V&V?	What this note records
NEG execution sheets	SKT-TC-NEG-001	After fault tests	Some behaviours exist; expected messages not frozen
SEC execution sheets	SKT-TC-SEC-001	After security freeze	001 / 004 / 009 exist; kiosk / at-rest / SBOM do not

3. SKT-TC-HR-001 — Phase 1 heart-rate mapping

Aligned on both sides (code + SRS/SVVR): contiguous HR bands. No hole at 84.5 or 94.5.

`musicCategoryFromAvgHr` and SRS-ALG-001 / SVVR Phase 1 table:

HR (bpm)	Category
`null` / not finite / `<= 0`	Calming
`>= 95`	Grounding
`>= 85` and `< 95`	Calming
`>= 70` and `< 85`	Restorative
`< 70`	Uplifting

Examples: **84.5 Restorative**, **94.5 Calming**, **95 Grounding**.

Test ID	Title	As-built
TC-HR-001	Phase 1 HR category mapping	PASS
TC-HR-002	Phase 1 HR boundary handling	PASS

4. SKT-TC-HRV-001 — HRV algorithm verification

4.1 Formulas as implemented (`src/lib/hrv.ts`)

Not approved expected values. Independent calculation must use an agreed method, not this file as the answer key.

Item	As-built
RR keep range	300–2500 ms
Extra filter	IQR 1.5, then neighbour-average correction if `
Artifact %	`(range removed + IQR removed + corrected count) / initial count × 100`
RMSSD	`sqrt(sum(diff²) / (n − 1))` on corrected series
RMSSD min beats	**30** (`HRV_MIN_BEATS_RMSSD`)
SDNN	`sqrt(sum((RR − mean)²) / n)` — divisor **n**, not n−1
SDNN min beats	**60** (`HRV_MIN_BEATS_SDNN`)
pNN50	share of successive diffs `> 50` ms, same 30-beat floor as RMSSD
Rolling window	150 s default (`DEFAULT_HRV_CYCLE_SECONDS` in `rollingHrvWindow.ts`)
Window inclusion	timestamps `>= now − windowMs`
Live phase abort	artifact **> 10%** (`PHASE_ARTIFACT_ERROR_PCT`); UI warn path also uses **5%**
RMSSD % for music	`round(value × 100) / 100` (2 decimal places)

4.2 Phase 2 category as implemented (`getCategoryFromRmssd`)

RMSSD (ms)	Category the app uses
`null` / not finite	Calming
`<= 20`	Grounding
`<= 35`	Calming
`<= 55`	Restorative
`> 55`	Uplifting

Aligned on both sides (code + SRS-ALG-002 / SVVR):

RMSSD (ms)	Category
`<= 20`	Grounding
`> 20` and `<= 35`	Calming
`> 35` and `<= 55`	Restorative
`> 55`	Uplifting

Examples: **20.0 Grounding**, **20.01 Calming**. No hole at 20 / 35 / 55.

4.3 Switch-suppression as implemented (`musicCategoryFromRmssd`)

- First valid RMSSD: map category, allow ranking.
- Later: absolute RMSSD change `<= 20%` → keep previous category, do **not** reselect track.
- Change `> 20%` → remap category and allow ranking.

This matches the design intent (category change **and** ≥20% shift). Still not an executed PASS.

4.4 Per test ID

Test ID	Title	Word-pack gate	Actual / PASS?	As-built
TC-HRV-001	Known RR vector — RMSSD	BLOCKED until expected value + tolerance approved	**No**	Formula and 30-beat floor exist; need Dataset A + independent calc
TC-HRV-002	Known RR vector — SDNN	Same	**No**	Divisor n, 60-beat floor; need Dataset A + independent calc
TC-HRV-003	Window boundary	2.5 min inclusion rule not approved	**No**	150 s window; `t >= tMin`
TC-HRV-004	Insufficient data	General entry	Not until QA accepts criterion, then a run	RMSSD `null` if < 30 clean beats
TC-HRV-005	Artifact handling	General entry	Not until method is the approved one	Range + IQR + 25% correction; phase abort > 10%
TC-HRV-006	Missing data	General entry	Not until SRS wording signed	Invalid RMSSD → Calming or previous category
TC-HRV-007	Threshold below	Aligned ≤20 / ≤35 / ≤55	PASS	See §4.2
TC-HRV-008	Threshold equal	Equality uses ≤	PASS	20 Grounding, 35 Calming, 55 Restorative
TC-HRV-009	Threshold above	Aligned	PASS	20.01 Calming, 35.01 Restorative, 55.01 Uplifting
TC-HRV-010	Switch-suppression	General entry	Not until a controlled run	`<= 20%` keeps category
TC-HRV-011	Rounding / precision	Decimal precision not approved	**No**	Payload % uses 2 decimals; UI often `toFixed`

5. SKT-TC-BLE-001 — Polar H10

Product has pairing, RR stream, reconnect copy, sensor-off handling, and battery % (`BleProvider`, `ble.manager.ts`, `BleConnectPanel`).

Test ID	Title	Actual / PASS?	As-built / blocker
TC-BLE-001	Initial pairing	After Polar run on frozen build	Discover + connect exists
TC-BLE-002	Wrong nearby sensor	After run	User picks a device from the scan list
TC-BLE-003	RR stream acquisition	After run	RR intervals ingested with timestamps
TC-BLE-004	Temporary connection loss	**Blocked**	Timeout still `[X]` in SRS; app has HR grace / stale-packet timeout, not frozen
TC-BLE-005	Reconnection	After run	Reconnect path exists
TC-BLE-006	Sensor powered off	After run	Error state; do not use stale RR
TC-BLE-007	Low battery	After run (or N/A with QA note)	Battery Service % is shown
TC-BLE-008	Malformed / missing data	**Blocked**	Needs a qualified RR fault-injection harness — none in the Trial app
TC-BLE-009	Long session stability	After run	20-minute session exists
TC-BLE-010	Resume after interruption	After run	App interrupt / resume exists; confirm labelled behaviour

6. SKT-TC-E2E-001 — End to end

Test ID	Title	Actual / PASS?	As-built / blocker
TC-E2E-001	Authorized user starts app	After frozen build	OTP + 4-digit PIN
TC-E2E-002	Create / select patient	After run	Trial patients + session IDs
TC-E2E-003	Connect Polar H10	After run	Same as BLE-001/003
TC-E2E-004	Baseline / data-quality	After run	Resting baseline + artifact / missing-vitals gates
TC-E2E-005	HRV calculation	**Blocked**	Same as HRV-001/002 (no approved expected numbers)
TC-E2E-006	Therapy decision	PASS	Contiguous HR then RMSSD bands + 20% switch rule
TC-E2E-007	Audio playback	After run	Category-based streaming playback
TC-E2E-008	Mid-session signal loss	After run	Vitals-missing / BLE loss UI
TC-E2E-009	Session completion	After run	Default cap 20 minutes (`DEFAULT_SESSION_MAX_SECONDS`)
TC-E2E-010	Report / store result	**Blocked**	SRS requires AES-256 **local** session store + protocol offline/sync. App uses MMKV; session payload goes to Trial API over TLS. PIN storage mentions AES. That is not the labelled at-rest control.
TC-E2E-011	New patient after prior	After run	Session patient / music / discrete-band state is reset between sessions
TC-E2E-012	Application restart	After run	Confirm recovery against approved spec

7. SKT-TC-NEG-001 — Negative / fault

Expected messages are not frozen. Do not pre-fill PASS.

Test ID	Title	Product today
TC-NEG-001	App terminated during session	Partial data / restart path exists; recovery spec not frozen
TC-NEG-002	Tablet power loss	No specified safe-state in app
TC-NEG-003	Storage full	No dedicated warning / block
TC-NEG-004	Corrupt local record	No dedicated handler
TC-NEG-005	Bluetooth permission revoked	OS / BLE error path exists
TC-NEG-006	Audio device removed	Playback engine can pause / fail; message not frozen
TC-NEG-007	Rapid repeated taps	Session start should be single-flight; confirm on device
TC-NEG-008	Invalid patient fields	Form validation exists
TC-NEG-009	Date / time change	Not specified
TC-NEG-010	Unsupported OS	Not specified; production target is OnePlus Pad Go / Android 14

8. SKT-TC-SEC-001 — Cybersecurity

Aligns with `content/skitii-trial-tablet-security.md`: app lock is not device lock.

Test ID	Title	Product today
TC-SEC-001	Role / authentication	OTP + PIN; unauthorized user cannot use clinical functions
TC-SEC-002	Kiosk escape	**Not in the app.** Needs MDM Device Owner / lock-task on hospital units
TC-SEC-003	Data at rest	**Not AES-256 session store.** MMKV for PIN, drafts, tokens; clinical session records are server-side
TC-SEC-004	Data in transit	HTTPS to Trial / auth APIs; no certificate pinning in the app
TC-SEC-005	Update control	No labelled signed-update path
TC-SEC-006	Package integrity	Release APK can be hashed; QMS process not recorded here
TC-SEC-007	USB / peripheral	Debug USB used in lab; production must disable via MDM
TC-SEC-008	Audit / logging	Not a data-minimised security event log as specified
TC-SEC-009	Default credentials	No shared default admin password; OTP + PIN
TC-SEC-010	Dependency vulnerabilities	**Blocked** until production SBOM + qualified scanner + dispositions

9. OPEN items that still block a formal run (from SKT-HARM-001)

ID	Decision needed	Blocks
OPEN-01	Phase 1 HR bands aligned in code + SRS/SVVR (≥95 / ≥85 / ≥70)	Closed for 84.5 / 94.5 / 95
OPEN-02	Phase 2 RMSSD bands aligned in code + SRS/SVVR (≤20 / ≤35 / ≤55)	Closed for 20 / 35 / 55 ms
OPEN-03	Personal scoring parameters (α, bounds, exploration, min exposures)	SRS-ALG-004 / H-05
OPEN-04	Safe headphone volume limit	SRS-AUD-005
OPEN-05	Final tablet model + Android OS build	System validation (target today: OnePlus Pad Go OPD2304)
OPEN-06	Polar H10 firmware + BLE library version	Interoperability baseline
OPEN-07	Startup / connect / latency / storage numeric targets	Performance V&V
OPEN-08	Production crypto / key management / update signing	TC-SEC-003/005/006, TC-E2E-010
OPEN-09	Final SBOM + vulnerability scan	TC-SEC-010
OPEN-10	Formal IEC 62304 software safety class	Lifecycle tailoring
OPEN-11	Actual executed V&V evidence	Final SVVR / submission

10. Suggested order of work (engineering + QA)

1. PI / Quality freeze OPEN-01 and OPEN-02 in SRS + SDS (do not reverse-approve from this note).
2. Create Dataset A / B (known RR vectors) and **independent** RMSSD / SDNN expected values + tolerances.
3. Tag a release-candidate APK; record SHA-256, tablet ID, Polar H10 FW, `react-native-ble-plx` version.
4. Execute BLE (except 004 / 008), then E2E (except 005 / 006 / 010), then HRV-004/005/006/010, then NEG/SEC items that have a spec.
5. Index evidence in SKT-LOGS-001 Annexure S. Only then fill Word Actual / PASS cells and RTM Evidence.

Until step 1–3 exist, **do not update data inside SKT-TC-HR-001 or SKT-TC-HRV-001**.

11. Code pointers

Topic	File
Phase 1 HR / Phase 2 RMSSD / 20% switch	`Skitii-Trials-Yenepoya-HNF-APP/src/lib/musicAnalyzer/discreteBands.ts`
RMSSD / SDNN / artifact preprocess	`Skitii-Trials-Yenepoya-HNF-APP/src/lib/hrv.ts`
150 s window	`Skitii-Trials-Yenepoya-HNF-APP/src/lib/rollingHrvWindow.ts`
20-minute session cap	`Skitii-Trials-Yenepoya-HNF-APP/src/lib/sessionPhaseConstants.ts`
Phase artifact 10%	`Skitii-Trials-Yenepoya-HNF-APP/src/screens/Session/vitalsMissingNav.ts`
BLE + battery	`Skitii-Trials-Yenepoya-HNF-APP/src/hooks/ble.manager.ts`
Tablet kiosk gap	`Skitii-Docs-Generation/content/skitii-trial-tablet-security.md`